

Mid-Semester Presentation: Script and Question Bank

Agentic AI and RAG-Based Framework for Digital Forensics

How to use this: read the script aloud once or twice and put it in your own words; do not recite it. Times add up to about 16 minutes; the shorter route (about 11 minutes) is marked. If your panel is short on time, skip slides 5, 10 and 16 with one sentence each.

Part 1. Before you walk in

- Start the API: **python -m uvicorn api.main:app --port 8000**, open **http://localhost:8000/**, and send one warm-up text. The first request takes about 15 seconds while models load; after that 2 to 6 seconds.
- Check the three status lights in the page header (service, instruments, reviewer). "Model reviewer off" or "Not used" is expected: the free Gemini key has 20 requests a day and is used up. Say so calmly if asked (see Q&A 31).
- Have open: the slides, the portrait image you tested, **demo_samples/** (4 files), and the printed case record PDF as a backup if the demo fails.
- Smart App Control must stay OFF on the demo computer. Test with **python -c "import torch"** (prints nothing if fine).
- Suggested split for four people (change as you like): Slides 1-4 first speaker, 5-8 second, 9-11 and the demo third, 12-15 fourth.

Part 2. Slide-by-slide script

Slide 1: Title (0:30)

Good morning. We are presenting our mid-semester progress on "Agentic AI and RAG-Based Framework for Digital Forensics", guided by Dr. Anish Kumar Vishwakarma. Our team is Aditya Khatri, Abhay Kumar Kashyap, Devansh Singh Rawat and Vanshika Trikhya.

Slide 2: Content (0:30)

We will cover the problem, what existing work says, our objectives, how we organised the work, what is built, the results, and the conclusion. In one sentence: the framework already runs end to end on local models behind a web interface, six of our seven sub-objectives are complete, and we will be open about what is still weak.

Slide 3: Introduction (1:00)

Generative AI now produces fluent text and realistic faces very cheaply, so how something looks no longer tells you where it came from. A typical detector gives one probability and no evidence, and often fails on a new generator or a new data source. A forensic examiner needs three things: what was examined, how reliable the result is, and a clear signal when a person must decide. So our problem statement is: build an end-to-end framework that checks text and images with several different tools, combines them under clear rules, cites retrieved knowledge, and refers doubtful cases to a person. It applies to evidence screening, newsrooms, identity-fraud checks, media verification and academic screening.

Slide 4: Concepts and datasets (1:00)

On the left are the techniques we actually use: a fine-tuned DeBERTa text classifier, the Binoculars statistic, an EfficientNet-B4 face-swap detector with Grad-CAM, CLIP probes, calibration, retrieval-augmented generation, Reflexion and LangGraph. On the right are the datasets: HC3, MAGE and RAID for text, and WildDeepfake, FaceForensics++ and Celeb-DF for face swaps, plus two sets for synthetic faces and art. The important design point is the split.

We split text by source document and images by source video, and we kept whole generators, attack types and manipulation methods completely out of training, so our test scores measure generalisation and not memorisation.

Slide 5: Literature review, related work (1:00) (skippable)

We reviewed text benchmarks like RAID, zero-shot detectors like Binoculars, image benchmarks like FaceForensics++ and WildDeepfake, retrieval-augmented generation, and the recent ForenAgent work on agentic forensics. The common finding is that performance depends on the data distribution. That is why we test on held-out generators, attacks and datasets.

Slide 6: Inference and research gap (1:00)

Three conclusions: performance depends on the data; different signals complement each other; and an explanation is only useful if it points to evidence. The gap is that existing work addresses one detector, one modality or one reasoning method, while we want one workflow where a controller chooses tools, retrieved knowledge supports the result, and disagreement is checked before reporting. We saw the first conclusion ourselves: our first text model scored 1.00 on its own data and 0.66 on MAGE, and our first image model scored 0.97 on its own data and 0.52, which is chance, on FaceForensics++.

Slide 7: Objectives (1:00)

Our main objective is a framework that combines an agentic controller and retrieval-augmented generation to judge text and images, with explainable, evidence-backed reports. Of the seven sub-objectives, four are done as planned, and three are done with modifications that we explain on the next slides: the text detector, the image pipeline, and the prototype. Evaluation is partly done: nine held-out sets for the local tools are complete, and the language-model judge rows are pending.

Slide 8: Methodology (1:00)

We used an iterative, evaluation-driven method: build a tool, measure it on held-out data, diagnose the failure, fix it, and measure again. That loop found our two biggest problems. The six steps are on the left, and the timeline on the right shows phases A to F done and phase G, evaluation and documentation, in progress. We changed the plan in a few places and recorded each change with its reason in a decision log. We dropped the XGBoost model because the transformer trained on three corpora was far stronger. We used DeBERTa instead of RoBERTa. We used CLIP probes instead of frequency tools. And we built a custom web interface instead of React.

Slide 9: Architecture (1:30)

This is the system as built. A request enters the API, which validates it by content, not by file name. A LangGraph orchestrator routes it by type. For text, three tools run: the DeBERTa classifier, the Binoculars statistic, and an optional language-model judge. For images, a scope guard first checks whether the image is a face photo, because our deepfake detectors only know faces. Face photos go to EfficientNet-B4 with Grad-CAM and a CLIP face probe. Other images go to a general AI-image probe. Then the RAG layer retrieves cited reference notes, and the verifier fuses the calibrated scores, checks disagreement, and refers uncertain cases to a person. Every case is saved with a SHA-256 hash of the input.

Slide 10: Modules and deliverables (0:45) (skippable)

This table lists what is attained against what remains. Engineering, text agent, image agent, orchestration, RAG and the product are all working. Remaining items are mostly judge calibration, better transfer to unseen manipulation methods, and a larger knowledge base.

Slide 11: Interface and live demo (2:30)

This is the interface a reviewer sees. Let me show it live.

1. **Human text** (demo_samples/text_human_written.txt): result "Authentic". Point at the finding, the meter showing where the combined reading sits against the thresholds, and the instrument readings.
2. **AI text** (text_AI_generated.txt): result "Synthetic", confidence about 95 percent. Point at the cited reference material.
3. **Deepfake image** (image_deepfake.jpg): "Synthetic". Show the Grad-CAM figure: warm areas are where the network found evidence.
4. **Your own portrait** (image.png): show the side-by-side figure. If you know whether it is real or AI, press "Record determination" to show how a human correction feeds case memory.
5. Press **Print record** to show the one-page case record that can be saved as PDF.

Two honest points on the screen: the "Model reviewer" row says "Not used" because our free Gemini key allows only 20 calls a day, and the system works on the local instruments regardless. And earlier cases in the reference list are labelled unverified, because they are the system's own earlier outputs and only a human correction makes them verified.

If the demo fails: say "the models did not load on this machine", show the screenshots on this slide and the saved case record PDF, and carry on with the results slides. Do not try to debug live.

Slide 12: Results, text (1:15)

On five held-out text sets of 1,000 samples each, the full system reaches AUROC between 0.957 and 0.998, including a generator it never saw and attack types it never saw. At a strict one percent false-positive budget it still detects 58 to 61 percent of AI text on the hardest sets. The system refers 30 to 52 percent of cases to a person and is 92 to 99.6 percent accurate on the cases it decides. Two honest findings: the Binoculars probe adds no accuracy, so we capped its weight at 0.10 and keep it only as a second opinion; and a simple input normalisation step lifted the homoglyph attack from 0.72 to 0.99 without retraining.

Slide 13: Results, images (1:15)

On face-swap sets of 1,500 images each, the system reaches 0.976 to 0.999 AUROC on WildDeepfake, FaceForensics++ and Celeb-DF. But the last bar is the honest one: on manipulation methods it never saw, Face2Face and FaceShifter, AUROC is 0.794 and it catches only 13 percent of fakes at one percent false positives. Note that Celeb-DF and FaceForensics++ were in our training mix, with video-disjoint test sets, so only that last result is truly zero-shot. We tried self-blended images, an extra manipulation method and a CLIP-only probe; none closed the gap, and fusing CLIP with the CNN gave us plus 0.04.

Slide 14: Robustness and fixes (1:15)

To mimic social media, we degraded images with compression, downscaling and screenshot-like damage. Ranking quality mostly survives, but false alarms on real faces rise: on screenshot-like copies from 3 to 9 percent on Celeb-DF, 7 to 17 on WildDeepfake, and 15 to 49 on FaceForensics++. We fitted a second threshold on degraded calibration images only. It halves false alarms on Celeb-DF and WildDeepfake for 2 to 4 points of fakes caught, and helps only partly on FaceForensics++. On the right are the problems the evaluation exposed and how we fixed them, for example the 0.52 on FaceForensics++ that came from a preprocessing mismatch we found by inspecting the data.

Slide 15: Conclusion (1:15)

The system is built and tested end to end: 81 automated tests, continuous integration and a Docker build all pass. The main lesson is methodological: in-domain scores of 0.97 to 1.00 hid collapses on data from other sources, and diagnosing that was most of the work. Next we will calibrate and benchmark the language-model judges, detect re-uploaded images automatically, add a second unseen synthetic-face set and a non-native-English check, and finish the report. We state our limitations openly: unseen deepfake methods, re-uploaded images, and judges not yet benchmarked.

Slide 16: References (0:15)

The cited papers are listed here, from the synopsis and added during implementation.

Slide 17: Thank you (0:15)

Thank you. We are happy to take questions.

Part 3. Numbers to remember

Item	Value
Text AUROC, five held-out sets (1,000 each)	0.957 to 0.998 (MAGE 0.961, RAID all 0.985, unseen generator 0.957, unseen attacks 0.962, HC3 0.998)
Text, TPR at 1% false positives	0.58 (MAGE), 0.61 (unseen attacks), 0.81 (RAID all), 1.00 (HC3)
Image AUROC (1,500 each)	WildDeepfake 0.977, FF++ seen 0.976, Celeb-DF 0.999, FF++ unseen methods 0.794
Unseen methods, fakes caught at 1% FPR	13%
Referred to a person / accuracy when decided	29 to 52% referred (53% on unseen methods); on the rest 92 to 99.6% accurate (text and seen image sets), 76% on unseen methods
Early failures that we fixed	Text 0.66 on MAGE to 0.97; images 0.52 on FF++ to 0.98; homoglyph attack 0.72 to 0.99
Re-upload false alarms (screenshot-like)	Celeb-DF 9%, WildDeepfake 17%, FF++ 49%; with degraded threshold 4%, 8%, 37%
Training data for text	136,000 rows (HC3 24k, MAGE 60k, RAID 52k)
Models	DeBERTa-v3-base; EfficientNet-B4; CLIP ViT-L/14 probes; Qwen2.5-0.5B for Binoculars
Tests / knowledge base	81 tests; 21 knowledge-base chunks
Hardware / speed	RTX 5080 16 GB; 2 to 6 s per case, about 15 s for the first request
Judge status	Implemented and unit-tested; not benchmarked (free key: 20 calls a day); judge weight 0.3, not fitted

Part 4. Questions about the presentation (by slide)

Problem and motivation (slides 3 to 6)

1. Why is a single detector not enough?

A single detector gives a score with no evidence and no way to say it is unsure, and it tends to fail

on data unlike its training set. We saw this: 1.00 on its own data, 0.66 on MAGE. We use several independent tools, calibrate them, and refer cases where they cannot settle.

2. What is new in your work compared with existing detectors?

We do not claim a new base classifier. The contribution is the integrated, measured workflow: tool selection by a controller, calibrated fusion, cited retrieved evidence, disagreement checking and human referral, plus an evaluation designed around held-out generators, attacks and datasets. We also report negative results.

3. What does "agentic" mean here? Is it not just a pipeline?

The detectors are ordinary tools. The agentic part is the LangGraph controller that decides what happens per case: it routes by modality, uses a scope guard to choose tools for images, skips unusable tools, and, when weighted tools disagree and a judge is available, re-runs the judge once with the other tools' outputs (Reflexion) before deciding to refer. We are honest that the Reflexion loop is implemented and tested but not yet benchmarked.

4. Why text and images only, not video?

Those two let us build and evaluate the full architecture within the time. Video is listed as future scope, reusing the team's earlier spatio-temporal work.

5. Which of your literature sources matter most?

RAID [1] for showing detectors fail off-distribution, FaceForensics++ and WildDeepfake [4], [5] for the image benchmarks, Binoculars [3] which we implemented, and ForenAgent [7] for agentic tool use in forensics.

Datasets and method (slides 4, 7, 8)

6. How do you know there is no data leakage?

Text is split by source document (and by question for HC3); images by source video. Whole generators (Cohere) and attack types (paraphrase, synonym, homograph) never appear in training. The text preparation script has automatic overlap checks. We also retired two old test files because they overlapped the new training mix.

7. Why did you not report 99 percent accuracy?

Because that number is in-domain. HC3 gives AUROC 1.000 but it is an easy set. We report held-out and unseen results, which are the honest measure.

8. What is calibration and why do you need it?

Raw model outputs are not probabilities. We fit a temperature on a separate calibration split so that scores from different tools are comparable and can be combined. The thresholds and weights are fixed on that split before any test set is touched.

9. Why did you drop XGBoost?

The transformer trained on three corpora was far stronger and handcrafted stylometric features added little; keeping a weak tool needs a reason. The change is recorded in the decision log.

10. Why DeBERTa and not RoBERTa as in the synopsis?

The synopsis allowed RoBERTa-base or a comparable encoder; DeBERTa-v3-base is a strong encoder that fits the 16 GB GPU, and the real problem in the first model was the single training corpus, not the architecture. The big gain came from mixing HC3, MAGE and RAID.

11. Why did you change the training data after the first results?

The first text model memorised HC3's style (AUROC 0.66 on MAGE) and the first image model failed on FaceForensics++ (0.52) because of full frames versus face crops and a single source. Diagnosing that led to the mixed training data.

Architecture (slides 9 to 11)

12. What is the scope guard and why is it needed?

A zero-shot CLIP check that decides whether an image is a face photo. Our deepfake detectors

only know faces; without it, an AI illustration was labelled authentic in a live test. Non-face images go to a general AI-image probe instead.

13. What is Grad-CAM and can I trust it?

Grad-CAM highlights image regions that most influenced the network's score. It shows where the model looked, not proof of manipulation, so the interface says "indicative only".

14. What does RAG do here? Does it change the verdict?

It retrieves cited reference notes (literature, generator profiles, earlier cases) so a report can point to sources. By design it never changes the score; it only grounds the report and can add a referral flag when a human-verified similar case contradicts the verdict. It also keeps ablations clean.

15. How does fusion work?

Each usable tool contributes $\text{weight} = \text{base weight} \times \text{length factor} \times \text{confidence}$, with base weights from validation AUROC. The fused score is compared with two thresholds: above the upper one is synthetic, below the lower one authentic, in between uncertain.

16. When is a case referred to a human?

When fewer than two tools are usable, the fused score is in the uncertain band, the weighted tools still disagree, or a verified earlier case contradicts the verdict. One exception: for non-face images the general probe is the designated sole detector, so a confident verdict is not force-referred.

17. What is Reflexion?

When weighted tools disagree and a judge works, the judge is run once more with the other tools' outputs and the retrieved notes, then the verdict is recomputed. At most one retry. It is implemented and tested with mocks; we have not yet measured its effect.

18. Why LangGraph and ChromaDB?

LangGraph gives a typed shared state and explicit routing, which makes each stage testable. ChromaDB is simple, local and persistent, enough at our scale; we considered Qdrant for production.

19. How is evidence integrity handled?

Each case stores a SHA-256 hash of the input, tool verdicts, citations, model and calibration versions, and any human correction. The raw content is not stored in the record.

20. Is the system secure?

Inputs are validated by content, an optional access key and a rate limit are available, jobs run one at a time, and uploads can be deleted automatically after a chosen number of days. Secrets are never in the repository. We have not done a formal security audit or prompt-injection red-teaming beyond treating input as data.

21. Where does it run and what are the hardware needs?

On one RTX 5080 16 GB laptop GPU; 2 to 6 seconds per case after models load. It is also packaged in Docker; the container build is checked in our continuous integration.

Results (slides 12 to 14)

22. What do AUROC, TPR at 1% FPR and ECE mean?

AUROC is the probability that a random fake scores higher than a random real item (1.0 perfect, 0.5 chance). TPR at 1% FPR is the share of fakes caught while wrongly flagging only 1 percent of real items. ECE measures how far stated probabilities are from actual frequencies (lower is better).

23. Why is the unseen-method result so low?

The CNN learns the fingerprints of the methods it trained on. Face2Face is reenactment and FaceShifter is a different swap, so those fingerprints do not transfer. We tried self-blended images, an extra manipulation type and a CLIP-only probe; only CLIP fusion helped, by 0.04. We report it as an open weakness.

24. Binoculars is weak; why keep it?

On our calibration set it lowers fused AUROC at every weight (0.9947 without it). We keep it at weight 0.10 as an independent second opinion for disagreement detection; it costs about 0.003 AUROC. Its weight is capped by a documented rule in the calibration script.

25. 30 to 52 percent referred sounds like a lot. Is the system useful?

It is a trade-off we measure openly. Accuracy on the cases it decides is higher than overall accuracy on all nine held-out sets (for example MAGE 0.958 versus 0.924, unseen methods 0.759 versus 0.697), and 92 to 99.6 percent on the text and seen image sets, so referral concentrates the errors on the cases a person checks. The referral rate depends on the thresholds, which can be tuned for a different cost balance.

26. Are your Celeb-DF and FaceForensics++ results zero-shot?

No. Those datasets were in the training mix, with video-disjoint test sets, so the scores are per-dataset. The only zero-shot image test is Face2Face and FaceShifter, which is why we single it out.

27. Why do real faces get flagged after compression?

Compression and downscaling remove detail the detectors rely on, so real images start to look "different". We measured it, fitted a degraded-image threshold that halves false alarms on two sets, and plan automatic detection of re-uploads.

28. What happens with non-native English writing?

We did not measure this: no suitable dataset was available. Detectors are known to be biased against such writing, and our knowledge base says so. A dedicated audit is on the end-semester plan.

Judges, limitations, next steps (slides 14, 15)

29. Why is the LLM judge not benchmarked?

The Gemini key we have is free tier: 20 requests per day. A calibration needs hundreds of calls. We implemented and tested the judges, ran them on the four demo inputs, and set a conservative weight of 0.3. A full benchmark needs a billing-enabled key.

30. What did the judge actually do on your demo inputs?

The text judge scored the AI sample 0.72 and the human sample 0.05, correctly. The vision judge scored both images 0.05, so it missed the deepfake with 85 percent confidence. That is why we do not trust it alone and keep its weight low.

31. Why does the interface say "Model reviewer: Not used"?

Because the daily quota of the free key is used up. The system is designed to continue on the local tools and says so on the page. It is not an error.

32. What will you do in the second half?

Calibrate and benchmark the judges and measure Reflexion; detect re-uploaded images automatically; add a second unseen synthetic-face set and a non-native-English audit; retrain the CNN with synthetic faces; test frequency evidence; finish the report.

33. Is this admissible as legal evidence?

No. It is a screening and decision-support tool that gives probabilistic indications. Legal chain of custody, signed logs and validation under a standard are listed as future scope.

34. What if someone attacks your detector?

We tested homoglyph, synonym and paraphrase attacks on text (AUROC 0.96 to 0.99 after normalisation) and compression and resizing on images. Targeted adversarial attacks against our specific models were not tested, and we state that.

Part 5. Questions about the report (by chapter)

Chapter 1, Introduction

35. What problem does the report solve?

See Section 1.3: an end-to-end forensic workflow for text and images that combines several tools under explicit rules, retrieves supporting knowledge, and returns a confidence-aware, explainable verdict, flagging cases for a human.

36. Which sub-objectives are complete?

Table in Section 1.6: six of seven; three of those with modifications (text detector, image pipeline, prototype). Evaluation is complete for the local tools and pending for the judges.

37. What differs from your synopsis?

Table in Section 3.1: XGBoost removed, DeBERTa-v3-base on a three-corpus mix instead of RoBERTa, CLIP probes and a scope guard instead of frequency tools, EfficientNet only instead of Xception, a custom web interface instead of React, ChromaDB only instead of Qdrant. Each has a recorded reason.

Chapter 2, Literature review

38. What is the research gap?

Section 2.2: prior work addresses one detector, one modality or one reasoning method; we study one text-and-image workflow with controller-chosen tools, retrieved evidence and disagreement checking.

39. Explain Binoculars in a sentence.

It compares how surprised one language model is by a text (perplexity) with how much two models disagree on it (cross-perplexity); machine text tends to give a lower ratio [3].

40. How do RAID and FaceForensics++ differ?

RAID is a text benchmark varying generator, domain, attack and decoding; FaceForensics++ is an image benchmark of facial manipulations at several compression levels.

Chapter 3, Design

41. Explain Figure 3.1.

Web interface to API, then the LangGraph orchestrator routing to the text agent or image agent, then RAG, then the verifier, then the report and audit record. The verifier may loop once to a judge (Reflexion).

42. What is input normalisation?

Zero-width characters are removed, NFKC normalisation is applied, and look-alike Cyrillic or Greek letters are mapped to ASCII when the text is mostly Latin. It removed the homoglyph attack's effect (0.72 to 0.99).

43. Why are fusion weights 0.99, 0.10, 0.96, 0.78, 0.88 and 0.3?

Base weights are $2 \times (\text{validation AUROC} - 0.5)$, fitted on the calibration split. Binoculars is capped at 0.10 because it hurts the fused score; the general image probe's weight comes from its held-out AUROC (0.94); the two judges are set to 0.3 because they could not be calibrated.

44. How are the thresholds chosen?

From the calibration split: an F1-optimal cut, with the uncertain band set from precision and negative-predictive-value targets and at least 0.05 wide on each side, then frozen before testing.

Chapter 4, Implementation

45. What is in the repository and how do you ensure quality?

Section 4.1: api, agents, rag, evidence store, training, eval, web, tests. 81 automated tests run without models or keys; a GitHub Actions pipeline runs lint, tests, and builds and imports both Docker images.

46. How were the models trained?

Section 4.2: transfer learning, early stopping on validation AUROC, bf16/fp32 (fp16 gave NaN losses), class weights where needed, and strong augmentation for images. Splits by document or video.

Chapter 5, Results

47. Which is your best and worst result?

Best: Celeb-DF 0.999 and HC3 0.998 (both seen-style data). Worst: unseen manipulation methods 0.794 with 13 percent of fakes caught at 1 percent false positives.

48. How did you answer the five research questions?

Section 5.4: (1) fusion helped images (+0.04 on unseen methods), not text; (2) the scope guard shows a routing benefit; (3) RAG makes reports cited but its quality was not rated by humans; (4) the verifier concentrates errors, 92 to 99.6 percent accurate when decided; (5) 2 to 6 seconds per case.

49. What would change your results most?

More diverse manipulation methods and generators in training, calibrated judges, and detecting re-uploaded images before choosing a threshold.

Chapter 6 and honesty questions

50. What are the main limitations?

Section 5.6: unseen manipulation methods, false alarms on re-uploaded images, judges not benchmarked, writing styles not measured (non-native English), partial detection of AI-synthesised faces, and results being probabilistic, not proof.

51. Did you use AI tools to write the report or code?

Answer truthfully and consistently with your acknowledgement page: AI assistants were used in developing the code and drafting parts of the report; all experiments ran on our data and every number comes from saved result files in the repository.

52. What did each of you contribute?

Prepare this yourselves, one honest sentence each. The report does not state individual contributions, so the panel may ask.

53. Can we see the code and the results again?

Yes: github.com/Aditya-Khatri-7/Major-Project- (branch Aditya-Khatri and main); results are in eval/results.

Part 6. Do not bluff on these

- The judges have **not** been benchmarked, and Reflexion has **not** been measured live. Say so plainly.
- The system does **not** reliably catch new face-swap methods (0.794 AUROC, 13 percent at 1 percent false positives).
- Re-uploaded images raise false alarms; the degraded threshold is measured but **not** enabled by default.
- Non-native-English writing was **not** measured.
- Celeb-DF and FaceForensics++ results are **not** zero-shot.
- If you are asked something you do not know, say: "We have not measured that; here is how we would test it."